

WAJEEHA ASLAM

AI ENGINEER | RAG SYSTEMS & KNOWLEDGE GRAPHS | LLM APPLICATIONS

Lahore, Pakistan | +92 313 4601985 | wajeehaaslam597@gmail.com | linkedin.com/in/wajeeha-aslam | github.com/WajeehaAslam

PROFILE

AI Engineer with 1+ year of experience building production-grade LLM applications, Retrieval-Augmented Generation (RAG) systems, and knowledge-graph-powered legal AI. Skilled at bridging complex, unstructured data and intelligent insight through ontology-based GraphRAG, agentic workflows, Text-to-SQL pipelines, and LLM-based automation. Hands-on with LangChain, Azure OpenAI, Azure AI Search, Neo4j, Qdrant, FastAPI, and Docker, with a track record of taking systems from prototype to measurable accuracy gains in production.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, FastAPI, LangChain, LangGraph, SQL, Docker, MCP
- **LLM & AI:** RAG System Architecture, Agentic Workflows, Knowledge Graphs for RAG, Text-to-SQL Pipelines, LLM Inference Optimization, OpenAI API
- **Databases & Search:** Neo4j, Vector Databases (Qdrant, FAISS), Azure AI Search, MySQL/PostgreSQL, Hugging Face
- **Cloud & DevOps:** Azure OpenAI, Azure Container Apps, Azure Blob Storage, Observability (Prometheus & Grafana), Pydantic Logfire
- **Tools:** Git/GitHub, Postman, REST APIs, Linux/Bash, Streamlit

PROFESSIONAL EXPERIENCE

Associate AI Consultant (AI Engineer) — Digifloat

Jul 2025 – Sep 2026

Lahore, Pakistan

Delivered end-to-end LLM, RAG, and knowledge-graph solutions across legal-tech, government, banking, and enterprise client engagements, from architecture design through evaluation and Azure deployment.

Ontology-Based GraphRAG (OG-RAG) — LawGPT

- **Designed a legal ontology and GraphRAG pipeline** that transformed unstructured case law into a structured knowledge graph of cases, judgments, judges, lawyers, parties, courts, statutes, and journals, with relationships including cites, follows, overrules, and applies-statute.
- **Built the extraction-to-graph pipeline**, retrieving cases from an Azure AI Search index and populating a Neo4j knowledge graph (1.41M nodes, 4.44M relationships across 50K cases) via an ontology-driven extraction script.
- **Developed an intelligent query router** combining Text-to-Cypher, vector search, and full-text search to answer structured, keyword, and semantic legal queries with citations.
- **Benchmarked OG-RAG against LawGPT RAG 1.0 on 50K legal cases**, cutting the evaluation error rate from 50% to 5%, sharply reducing hallucination risk, and unlocking corpus-wide analytical and multi-hop reasoning queries that plain RAG could not answer.

LawGPT Pakistan — Case Linking & Historization

- **Planned and designed a case-linking system** using title/party-name matching, legal keyword and tag matching, similar statutes, and cited precedents to surface related legal cases.
- **Designed a case-tagging approach and prompts** to extract legal tags from cases in support of automated case linking, plus a case-historization design to maintain full audit trails across case changes.
- **Delivered UI mockups and flow diagrams** for the case-linking and historization features ahead of implementation.

FAB — Policy Q&A Assistant (RAG)

- **Built an end-to-end Policy Q&A Assistant** on Azure OpenAI, Azure AI Search, LangChain, Azure Blob Storage, and Streamlit, helping employees query 50+ page policy PDFs instantly instead of manually searching them.
- **Implemented contextual-compression reranking, conversational query rewriting, and citation-aware answers** referencing the exact source PDF and page number, plus in-app document browsing and download.
- **Achieved 100% accuracy across 36/36 evaluation test cases** with average response latency of 6–9 seconds.

HRMS QA — RAG & Tool-Augmented Agent

- **Developed a RAG and tool-augmented agent** for HRMS-related queries, adding multi-layer guardrails for response safety and structured citation traceability.
- **Applied contextual compression to sharpen retrieval precision**, achieving an average response time of ~7.68 seconds and cutting manual information-retrieval effort; deployed on Azure Container Apps.

D2D — Data-to-Decisions (Text-to-SQL)

- **Built a reasoning-based Text-to-SQL pipeline** that plans against the database schema before generating and executing SQL against Dremio, plus a self-correction workflow that feeds execution results and reasoning back to the LLM with semantic validation and up to 3 automatic retries.
- **Improved benchmark accuracy from 90% (V1) to 94% (V2)**, and prepared and ingested datasets across banking, healthcare, clinical, procurement, pharmacy, and legal domains for client demos and pipeline testing.

LawGPT KSA — Legal Chat

- **Developed an AI-powered legal chat system** for Saudi Arabian laws and cases supporting Arabic and English queries, covering 4 legal PDFs (Labor, Basic Law of Governance, Traffic, Sharia Litigation) and 50 KSA MOJ cases.
- **Used Qdrant as the vector store with Hugging Face MiniLM-L12 embeddings** for semantic search, reaching ~90% response accuracy at ~10-second average response time.

TSG GPT — Document Drafting

- **Built an AI-driven document-drafting pipeline** that classifies requests, collects and validates structured inputs across 8 business dimensions (drivers, stakeholders, benefits, strategic alignment, timeline, budget, risk), and auto-generates complete drafts from either manual entry or uploaded documents.

RTA Mehboob — LLM Evaluation

- **Built an LLM-based evaluation pipeline** to classify generated responses as correct or incorrect across RTA services, creating a benchmark dataset covering 119 services with 5 questions each.

ADDITIONAL EXPERIENCE

- **Fellow, ACM-UET** — Worked on cloud infrastructure and data-management workflows supporting large-scale analytics on GCP.
- **Intern, Interns Pakistan** — Built web projects using HTML, CSS, JavaScript, PHP, and WordPress.
- **Intern, MLabs** — Created pixel-art game assets and collaborated with a development team to integrate them into gameplay.

EDUCATION

Bachelor's in Artificial Intelligence — COMSATS University Islamabad, Lahore Campus
CGPA: 3.13 | Feb 2023 – Present

LEADERSHIP & ACTIVITIES

- **Robotics and Automation Society (RAS), COMSATS Lahore** — Organized and managed society events (Feb 2024 – Present).
- **Lead 3 Interns at Digifloat.**